SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE CODE: MLA 02 COURSE NAME: Fundamentals Of Machine Learning

COURSE OUTCOME COVERED

CO4: *Develop probabilistic graphical models for performing inference, learning patterns, and analyzing sequential data.(BL4)*

Assessment Tool Weightage: 20%

QUESTIONS: 3 Total Marks:50 Annexure A

Industry Problem Solving Task

Code of Conduct:

I B.Harsha Vardhan(192425201) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Industry Problem Solving Task

1. Fraud Detection Using Bayesian Networks (20 Marks)

Industry Scenario
A digital banking organization aims to detect fraudulent credit card transactions. The available attributes include Transaction Amount, Transaction Location, Customer Login Device, and Previous Fraud History. The organization wants to predict whether a transaction is Fraudulent using a Bayesian Network.
Task

Develop a Python program to:

1. Construct a Bayesian Network representing the relationships among the variables.
2. Define the Conditional Probability Tables (CPTs).
3. Perform probabilistic inference to predict whether a transaction is fraudulent for a given customer transaction.
4. Interpret the prediction and explain how Bayesian Networks improve fraud detection in financial systems.

Fraud Detection Using Bayesian Networks

1. Construct Bayesian Network:
A Bayesian Network represents the probabilistic relationships between variables using a directed graph. For fraud detection, the network can contain:

Transaction Amount → Fraudulent
Transaction Location → Fraudulent
Login Device → Fraudulent
Previous Fraud History → Fraudulent

These variables influence the probability that a transaction is fraudulent.

2. Define CPTs:
Conditional Probability Tables (CPTs) contain the probability of each variable based on its parent variables. For example, the probability of fraud can be defined for combinations such as:

- High transaction amount
- Unknown location
- New login device
- Previous fraud history

A higher-risk combination receives a higher probability of fraud.

3. Probabilistic Inference:P(A|B) = \frac{P(A \cap B)}{P(B)}

4. Interpretation and Benefits:
If the calculated probability of fraud is high, the bank can flag the transaction for additional verification. Bayesian Networks improve fraud detection because they:

- Handle uncertainty effectively.
- Combine multiple factors.
- Provide probability-based predictions.
- Help identify relationships between transaction attributes.

- Support faster and more informed financial decisions.

Conclusion:
A Bayesian Network provides a systematic way to combine transaction information and previous fraud patterns to estimate the probability of fraud and help financial organizations detect suspicious transactions.

2. Predictive Machine Maintenance Using Hidden Markov Models (20 Marks)

Industry Scenario
A manufacturing company continuously monitors industrial machines using sensors that record Temperature, Vibration, and Motor Noise. The actual machine condition (Healthy, Warning, or Failure) is hidden. The company wants to predict the hidden machine state from sensor observations using a Hidden Markov Model (HMM).
Task
Develop a Python program to:

1. Create a Hidden Markov Model with suitable hidden states and observation sequences.
2. Define transition and emission probabilities.
3. Predict the hidden operational state of the machine for a sequence of sensor observations.
4. Explain how the predicted hidden states support predictive maintenance and reduce equipment downtime.

Ans: Predictive Machine Maintenance Using Hidden Markov Models

1. Create Hidden Markov Model:
An HMM is used because the actual machine condition cannot be directly observed. The hidden states are:

- **Healthy**
- **Warning**
- **Failure**

The observations are sensor readings such as **Temperature, Vibration, and Motor Noise**.

2. Transition and Emission Probabilities:

- **Transition probability** represents the probability of a machine moving from one state to another, such as Healthy → Warning or Warning → Failure.
- **Emission probability** represents the probability of observing a particular sensor condition given a hidden machine state.

For example, high temperature and high vibration are more likely to be emitted by a machine in the **Failure** state.

3. Predict Hidden Operational State:
The sequence of sensor observations is given to the HMM. Using the transition and emission probabilities, the model estimates the most likely sequence of hidden states.

Example:

Sensor Observations → HMM → Healthy → Healthy → Warning → Failure

The **Viterbi algorithm** can be used to find the most probable hidden-state sequence.

4. Benefits for Predictive Maintenance:
The predicted states help engineers identify machines that are moving from **Healthy → Warning → Failure**. Maintenance can be scheduled before a serious breakdown occurs.

This helps to:

- Reduce unexpected equipment failures.
- Reduce machine downtime.
- Plan maintenance in advance.
- Reduce repair costs.
- Improve machine reliability and productivity.

Conclusion:
HMM helps manufacturing companies estimate hidden machine conditions from sensor data and take preventive action before equipment failure occurs.

3. Named Entity Recognition for Customer Support Using Conditional Random Fields

Industry Scenario
A customer support company receives thousands of emails every day. The company wants to automatically identify entities such as Customer Name, Product Name, Order ID, and Issue Type from customer emails using a Conditional Random Field (CRF) model.
Task
Develop a Python program to:
1. Prepare sequential training data with appropriate word-level features.
2. Train a Conditional Random Field (CRF) model.
3. Predict entity labels for a new customer support email.
4. Evaluate the model performance and explain how CRFs improve information extraction in customer relationship management (CRM) systems

Ans:

Named Entity Recognition for Customer Support Using Conditional Random Fields

1. Prepare Training Data:
Customer emails are converted into sequences of words. Each word is assigned a label such as:

- **Customer Name**
- **Product Name**
- **Order ID**
- **Issue Type**
- **O (Other)**

Word-level features can include the word itself, first/last characters, capitalization, word position, and surrounding words.

2. Train CRF Model:
A **Conditional Random Field (CRF)** is trained using the prepared word sequences and their corresponding entity labels. CRF considers the relationship between neighboring labels, making it suitable for sequential text data.

3. Predict Entities:
When a new customer email is provided, the trained CRF model analyzes the words and predicts their entity labels.

Example:
"John ordered a Samsung Galaxy S25, Order ID 45821, but the screen is damaged."

Prediction:

- **John → Customer Name**
- **Samsung Galaxy S25 → Product Name**
- **45821 → Order ID**
- **screen is damaged → Issue Type**

4. Evaluation and CRM Benefits:
The model can be evaluated using **Precision, Recall, F1-score, and Accuracy**.

CRFs improve CRM systems by:

- Automatically extracting important customer information.
- Reducing manual data entry.
- Processing large numbers of emails quickly.
- Improving customer query classification.

- Helping support agents respond faster.

Conclusion:
CRFs effectively identify entities by considering both individual words and their surrounding context, making them useful for automated information extraction from customer support emails.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis

Professionalism	5%	Highly professional, well documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation